DE PROJECT TRACKER · SPRINT PLAN · GENERATED 12-09-26

E-Commerce Data Platform — Sprint Plan

Nine sprints, 14-08-26 to 04-11-26. Sprints 1–4 are recorded as delivered; sprint 5 is in progress; sprints 6–9 are planned at task grain so each row has an unambiguous done condition.

43	18	2	23	42%
DELIVERABLE ROWS	COMPLETED	IN PROGRESS	PLANNING	PROGRESS

TENTATIVE DELIVERABLES	MODULE_START	MODULE_END	STATUS	RESPONSIBLE	PRIORITY	REMARKS
Sprint 1 Foundation 14-08-26 → 16-08-26						2/2 complete
Problem Statement finalization	14-08-26	16-08-26	COMPLETED	All	HIGH	Finalized problem statement, project scope and initial architecture
Dataset Selection	16-08-26	16-08-26	COMPLETED	All	MEDIUM	Olist selected; 9 CSVs, 1,550,922 rows; data sources, formats and quality reviewed
Sprint 2 Technical analysis of existing market solutions 21-08-26 → 23-08-26						2/2 complete
Technical report (Lab 2)	21-08-26	23-08-26	COMPLETED	All	HIGH	Reviewed Shopify, BigQuery, Databricks, inventory tools. REWORK: in-text citations and IEEE refs required -> Sprint 5
Module architecture and dependency mapping	22-08-26	23-08-26	COMPLETED	All	HIGH	Modules 1.x-2.x defined. REWORK: reviewer asked for task-grain decomposition -> Sprint 5
Sprint 3 Data Engineering Architecture and Requirements 24-08-26 → 03-09-26						2/2 complete
Final Lab 3 report	24-08-26	31-08-26	COMPLETED	All	HIGH	Batch pipeline design: Airflow, Python, storage, Spark, dbt, BigQuery, NetworkX, FastAPI
Project Tracker requirements (FR/NFR + coding standards)	31-08-26	03-09-26	COMPLETED	All	HIGH	10 FR, 9 NFR, 19 coding rules. REWORK: semantic codes required -> Sprint 5
Sprint 4 Data Foundation 04-09-26 → 12-09-26						7/8 complete
Source identification and raw preservation (1.1-1.2)	04-09-26	05-09-26	COMPLETED	All	HIGH	6 table contracts declared with columns, keys, dtypes and quirk notes; source directory kept read-only
Data ingestion and manifest (1.3-1.5)	05-09-26	07-09-26	COMPLETED	All	HIGH	347,649 rows landed byte-identical; SHA-256 manifest; idempotency proven across reruns
Storage layer, dual backend (2.6)	07-09-26	08-09-26	COMPLETED	All	HIGH	StorageBackend interface with local and MinIO implementations; identical key layout verified on both
Data cleaning and validation (2.1-2.5)	08-09-26	10-09-26	COMPLETED	All	HIGH	18 quality gates; 112,650 unit rows -> 102,425 order lines; revenue invariant R$ 13,591,643.70 preserved
Analytical data preparation (2.7)	10-09-26	11-09-26	IN PROGRESS	All	MEDIUM	order_products and product_cooccurrence built; dbt models not started -> CARRY-OVER to Sprint 6

TENTATIVE DELIVERABLES	MODULE_START	MODULE_END	STATUS	RESPONSIBLE	PRIORITY	REMARKS
Order, product, seller and relationship analysis (2.8-2.11)	10-09-26	11-09-26	COMPLETED	All	MEDIUM	Revenue R$ 13.07M, AOV R$ 136.91, 4,058 pairs, 14.8% graph coverage. Ranking views deferred
Initial visualization (2.12)	11-09-26	12-09-26	COMPLETED	All	MEDIUM	5 figures on one visual system, written from profile.json so prose and charts cannot diverge
Coding standards and toolchain	04-09-26	12-09-26	COMPLETED	All	HIGH	19/19 tracker rules documented and enforced by make check; 24 unit tests; clean_all refactored 105 -> 29 lines
Sprint 5 Review Remediation and Graph Parameterisation	13-09-26 → 19-09-26					5/7 complete
In-text citations and IEEE references (FB-1, FB-2)	13-09-26	14-09-26	IN PROGRESS	All	HIGH	9 anchor positions mapped; 2 entries need author lists from source; Stocky marketplace URL replaced; Kafka claim must be sourced or dropped
Market analysis to architecture traceability (FB-3)	14-09-26	15-09-26	COMPLETED	All	HIGH	11-row table mapping each market finding to one named architecture decision and the artefact holding it
Product graph parameter study P1-P7 (FB-4)	13-09-26	16-09-26	COMPLETED	All	HIGH	3 assumptions overturned. Support rule co>=2 and pop>=5; tie-break (order_count, lift, product_id) cut ambiguity 62% -> 2%
Semantic FR/NFR recoding (FB-5)	16-09-26	17-09-26	COMPLETED	All	MEDIUM	22 requirements recoded by concern; 3 new requirements added; FR-STO-01 reworded to end the MinIO contradiction
Task-grain module decomposition (FB-6)	17-09-26	18-09-26	COMPLETED	All	HIGH	Exposed a real dependency inversion: synthetic inventory is a source and must precede the dbt models
Essence card assessments (FB-7)	18-09-26	19-09-26	COMPLETED	All	MEDIUM	Two dated assessments with evidence; Requirements stuck at Bounded pending the document reconciliation below
Reconcile the five design documents	18-09-26	19-09-26	PLANNING	All	HIGH	Storage said MinIO in 2 docs and local filesystem in 2; NFR-LIN-02 names Spark, not implemented. Fixing both advances Requirements to Coherent
Sprint 6 Synthetic Inventory and Warehouse Modelling	20-09-26 → 30-09-26					0/7 complete
Per-pair sales velocity in the declared window (3.1.1)	20-09-26	20-09-26	PLANNING	All	HIGH	Extract profile_velocity into a reusable module; window 2017-02-01 to 2018-08-31 as a named constant
Synthetic inventory generator (3.1.2-3.1.4)	21-09-26	23-09-26	PLANNING	All	HIGH	current_quantity as N days of cover derived from each pair's own velocity, so the restock rule can actually fire
Inventory contract and ingestion (3.1.5)	23-09-26	24-09-26	PLANNING	All	HIGH	34,448 rows at seller_id + product_id grain; labelled synthetic throughout
dbt staging models (3.2.1)	24-09-26	25-09-26	PLANNING	All	HIGH	Over processed Parquet; lowercase snake_case, named CTEs, no SELECT *
Dimensions with uniqueness tests (3.2.2)	26-09-26	27-09-26	PLANNING	All	HIGH	dim_products, dim_customers, dim_sellers, dim_dates. Date spine must cover 140 calendar days with zero orders
Facts with grain tests (3.2.3)	28-09-26	29-09-26	PLANNING	All	HIGH	fct_orders at order-product-line grain with derived quantity; grain test per fact table

TENTATIVE DELIVERABLES	MODULE_START	MODULE_END	STATUS	RESPONSIBLE	PRIORITY	REMARKS
BigQuery load and validation	29-09-26	30-09-26	PLANNING	All	HIGH	Verify sandbox default table expiry before committing the warehouse to it
Sprint 7 Product Graph and API 01-10-26 → 12-10-26						0/6 complete
fct_product_cooccurrence with lift and confidence	01-10-26	03-10-26	PLANNING	All	HIGH	Adds lift, conf_a_to_b, conf_b_to_a so the undirected store keeps the asymmetry (P3, P6)
Two-tier graph build	03-10-26	06-10-26	PLANNING	All	HIGH	Product tier serves 214 products; category tier covers 98.2% of catalogue as fallback (FR-GRF-03)
Deterministic neighbour ranking	06-10-26	07-10-26	PLANNING	All	HIGH	Order by (order_count, lift, product_id) per P5; satisfies NFR-GRF-02 and makes API output stable
Holdout evaluation harness	07-10-26	09-10-26	PLANNING	All	HIGH	2018-06-01 split; report coverage and hit rate as dbt tests so a regression fails CI, not the demo (NFR-GRF-03)
FastAPI endpoints	09-10-26	11-10-26	PLANNING	All	HIGH	/health, /products/{id}, /products/{id}/recommendations, POST /search; graph loaded once at startup
Relationship-aware search	11-10-26	12-10-26	PLANNING	All	MEDIUM	Category and attribute match only: the products table has no name or description text
Sprint 8 Inventory Analytics and Dashboard 13-10-26 → 24-10-26						0/6 complete
Velocity and days_of_inventory models	13-10-26	15-10-26	PLANNING	All	HIGH	Denominator fixed to the declared window and published in the model so the rate is reproducible
Restock priority rules	15-10-26	17-10-26	PLANNING	All	HIGH	Compare against reorder_point AND lead_time_days; zero-sales pairs excluded safely (NFR-INV-01)
Dashboard: Overview and Sales	18-10-26	20-10-26	PLANNING	All	MEDIUM	Warehouse-derived metrics only, so dashboard numbers cannot drift from SQL definitions
Dashboard: Inventory and Restocking	20-10-26	22-10-26	PLANNING	All	MEDIUM	Must use real velocity figures; the 6.0 units/day illustration is 4.5x the observed maximum
Dashboard: Product Insights and Operations	22-10-26	23-10-26	PLANNING	All	LOW	Requires reviews promoted from Optional to Must; 99.2% of orders carry a review score
API and dashboard test pass	23-10-26	24-10-26	PLANNING	All	HIGH	Status codes, response schemas, and displayed metrics reconciled against warehouse queries
Sprint 9 Orchestration and Integration 25-10-26 → 04-11-26						0/3 complete
Airflow DAG with fan-out	25-10-26	28-10-26	PLANNING	All	HIGH	Fan out after validate_models: graph and inventory branches are independent and must not run in series
End-to-end pipeline run	29-10-26	31-10-26	PLANNING	All	HIGH	One trigger produces every downstream artefact; tasks exchange locations, never DataFrames
Final report and demo	01-11-26	04-11-26	PLANNING	All	HIGH	13-step demo flow; lead with the coverage finding rather than a single working product

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How this maps onto the tracker sheet. Row 4 headers are the column contract: the sprint number and its start/end dates go on the first row of each sprint group only (columns A–C); every deliverable row fills D–J. Module_Start and Module_End are per-deliverable dates inside the sprint window — they are what feeds the date-wise progress map noted in cell E2, so leaving them blank disables it.

- **Status dropdown** currently offers only “Planning”. Extend the data validation to Planning / In Progress / Completed / Blocked / Carried Over, or the column cannot express a finished sprint.
- **Priority** must actually differentiate. Every seeded row was High, which carries no information; this plan uses High for path-critical work, Medium for work that can slip a sprint, Low for the one dashboard section that depends on a scope change.
- **Column K is instruction text** from the template, not a data column. Record real remarks in J and leave K as guidance.
- **For the KPIs column K asks for** (turnaround_time, defect rate, burndown, burnup), append status-change dates to the J remark as sub-fields, e.g. *“Planning 20-09 | In Progress 21-09 | Completed 23-09”*. Without those timestamps none of those KPIs are computable.
- **Project-level cells still to fill:** C2 Project_End_Date = 04-11-26, D2 Lead = a named person, F2 Progress (%) = 42%.
- **Date conflict to resolve:** sprint 3 ends 03-09-26, but the sprint progress note claims a window of 31-08-26 to 11-09-26, which overlaps it. Sprint 4 is set here to 04-09-26 → 12-09-26; the progress note should be re-dated to match, which also makes its completion claims true.